

Jon Morgan

Technical Director – Built Environment & Digital Integration

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"As a Principal Engineer, my work sits at the intersection of cross-disciplinary expertise, spanning sustainable design, building performance simulation, and software automation, complemented by a practical background managing technical delivery. I focus on designing bespoke data pipelines that connect physical building systems to operational performance and decarbonisation pathways, ensuring automation enhances productivity while delivering reliable, verifiable outputs. This combination of hands-on technical depth and operational experience helps me act as a practical bridge, translating complex engineering data into clear, transparent, and collaborative workflows that support teams and senior leadership."

Core Capabilities

- **Data Engineering & Automation:** Designing custom, reproducible data pipelines using Python to streamline complex technical workflows, automate data processing, and eliminate manual entry silos.
- **Artificial Intelligence:** Leveraging advanced AI tools—including Gemini, DeepSeek, and custom Gems—within development workflows to accelerate coding productivity, while designing practical system concepts like RAG-based architectures to unlock team knowledge bases.
- **Building Performance Simulation:** Extensive expertise in building services engineering, sustainable design, thermal comfort analysis, and energy efficiency profiling across property portfolios.
- **Systems Integration:** Bridging the gap between software infrastructure and physical assets, translating detailed performance metrics into clear, high-value strategic insights for stakeholders.

Professional Experience

Co-Perform Pty Ltd

Jul 2018 – Present

Co-Founder & Technical Director

Co-founded and scaled this specialised built environment consultancy. Direct firm-wide technical strategy, software infrastructure deployments, and commercial strategy while maintaining daily collaborative oversight of our operational pipelines, financials, and project delivery.

- **Strategic Leadership & Commercial Oversight:** Directed our corporate rebranding, messaging strategy, and a six-month strategic business review, maintaining close daily collaboration with my partner on firm-wide financials, cash position, proposals, invoicing, and work-in-progress (WIP).
- **Digital Operations & Systems Refactoring:** Exercised total organisational control over digital infrastructure; recently led a collaborative, comprehensive review and refactoring of all internal systems with a digital transformation specialist to improve system resilience, maximise uptime, and identify operational offshoring and outsourcing opportunities.
- **Municipal Policy Strategy & Technical Evidence:** Developed the technical evidence base and policy strategy supporting the City of Melbourne Sustainable Building Policy, in partnership with Nation Partners. Focussed on aligning municipal assets with Green Star pathways and identifying existing infrastructure to serve as extreme-heat cool centres.
- **Systems Engineering & DevOps:** Designed database-integrated data pipelines using Python to automate complex mathematical calculations and reporting; managed an AI-assisted DevOps repository to orchestrate cloud software infrastructure, continuous deployment, and testing.

- **Local AI Workspace & Prototyping:** Configured dedicated local LLM deployment environments (Ollama/DeepSeek) on personal GPU workstation infrastructure to securely execute rapid internal system refactoring and code generation pipelines.
- **R&D Tax Incentive Preparation:** Prepared all technical responses, scoped eligible research activities, mapped project hours, and compiled supporting evidence for annual federal R&D tax offset submissions, handed over to our accountant for final compilation and lodgement.
- **Existing Building Performance & NABERS Support:** Analysed upgrade scenarios for a major Spring Street high-rise, delivered multiple NABERS Commitment Agreement feasibility studies for large-scale commercial buildings, and compiled Building Management System (BMS) data logs to support the formal rating assessment for 101 Collins Street.
- **Building Performance & Climate Resilience:** Delivered energy, thermal, and daylight simulations for Sustainability Management Plans (SMPs) across industrial, commercial, residential, and community developments to secure municipal planning approvals while using future climate scenarios to stress-test comfort and mitigate heatwave risks.
- **Embodied Carbon & Data Integration:** Managed carbon, daylight, and energy calculations for a major 6-star Green Star healthcare project, developing a custom utility to parse contractor Bill of Materials (BoM) data for direct carbon accounting import.
- **Disaster Recovery & Redesign:** Rapidly restored operations within 48 hours following a global hosting outage by rebuilding the corporate web presence using Astro.

Moreland Energy Foundation Ltd & Visergy

Sep 2014 – Jul 2018

Senior Consultant (MEFL) / Principal & Sole Trader (Visergy)

Delivered high-level energy efficiency engineering, regional decarbonisation models, and performance-based design simulations for municipal portfolios and commercial real estate.

- **Sole Trader Operations (Visergy):** Established and managed an independent consultancy, directing all operations, proposal development, project delivery, invoicing, and client relationships.
- **Solar Feasibility & Predictive Tooling:** Created a predictive modelling tool to right-size solar installations, accelerating early-stage design feasibility and concept planning.
- **Climate-Resilient Public Housing:** Collaborated with DesignInc on public housing prototypes engineered for future climate scenarios, analysing comfort and overheating risks to protect vulnerable occupants.
- **Portfolio Performance & Approvals:** Modelled energy, daylight, and thermal performance across diverse asset classes to support town planning, Sustainability Management Plans (SMPs), and public housing retrofit decisions.
- **Regional Emissions Modelling & Optimization:** Conducted mathematical optimisation and analysed cross-sectoral regional datasets to build carbon inventories for the Hepburn Shire Net-Zero transition plan, translating complex interval data and physical metrics into clear visual models.

Arup (Melbourne)

Sep 2005 – Aug 2014

Leader of the Australasian Building Physics Team (2009 – 2014) • Senior Consulting Engineer (2005 – 2009)

Joined as a Senior Consulting Engineer, progressing to lead the regional Building Physics team for my final five years. Directed regional analytical capacity, workflows, hardware integration, and detailed computational modelling for one of the world's premier engineering institutions.

- **Graduate Mentorship & Technical Training:** Mentored building performance graduates and delivered structured training for engineering rotations. Participated directly in regional graduate recruitment.
- **Interdisciplinary Performance Forecasting:** Coordinated complex design parameters across services, facades, and acoustics, using future climate scenarios in dynamic thermal models to forecast long-term performance.

- **Building Performance & Quality Assurance:** Standardised simulation workflows across offices in Australasia, ensuring technical rigour, auditable results, and quality control on dynamic compliance calculations.
- **Computational Platform & Tool Integration:** Maintained an internal compute platform to centralise calculations, writing custom integration scripts to link airflow and daylight simulations for complex designs.

Arup (UK) & BMT Fluid Mechanics

Jun 2002 – Sep 2005

Consulting Engineer / Analyst

Delivered advanced airflow simulations, cladding pressure tests, facade load calculations, and process automation across commercial real estate, sports stadia, and offshore energy operations.

- **Built Environment Aerodynamics (BMT):** Used physical testing and simulation tools to conduct wind microclimate assessments, cladding pressure tests, and facade load calculations for complex tall buildings.
- **Systems & Modelling Automation:** Managed high-performance compute clusters and wrote custom 3D scripts to automate spatial modelling for design assessment tools.

Defence Science and Technology Organisation (DSTO)

1996 – 2000

Research & Development Systems Engineer

Executed advanced aerodynamic, hydrodynamic, and acoustic performance research for military aircraft, weapons systems, and submarines.

- **Aerospace & Defence Systems Engineering:** Conducted aerodynamic and hydrodynamic performance research for military aircraft (F/A-18 and F-111) and Collins-class submarines. Developed mathematical optimisation and fault-detection pipelines to diagnose jet engine degradation from physical test-bed logs.

Education

Professional Credentials & Industry Accreditations

- **Accredited Assessor – NABERS Upfront Carbon** (Embodied carbon assessment for materials, structure, and supply-chain metrics).
- **Energy & Sustainability Hub Advisory Board Member** – Building Designers Association of Victoria (2015 – 2018).

RMIT University

2009 – 2014

Master of Urban Planning and Environment

Advanced postgraduate studies focussing on urban environmental policy, microclimate resilience, sustainable infrastructure, and quantitative resource modelling.

RMIT University

1991 – 1995

Bachelor of Engineering (B.Eng.), Aerospace

Rigorous four-year engineering specialisation with deep focus on fluid dynamics and aerodynamics, structural mechanics, systems, control, and computational simulation.

| Strategy

My approach is built on a practical balance: the engineering background to analyse how buildings perform across their entire lifecycle—from initial design and construction through to daily facilities operation—paired with a strong awareness of the commercial drivers that shape these assets. Instead of relying on off-the-shelf software or siloed workflows, I create bespoke, integrated data pipelines that break down traditional discipline barriers to improve productivity. By translating complex building performance metrics into clear, structured insights, I ensure that technical workflows reliably support teams in optimising property portfolios for energy efficiency and long-term value. Combining sustainable design expertise with a proven ability to lead by example allows me to ensure that data systems serve as a dependable, high-value asset for a business delivering these outcomes.

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